

Week 18 – Backend Reliability & Stress Testing

Summary:

Week 18 focused on **stress testing** to verify backend reliability under heavy usage.

Key Activities:

- Simulated **multiple simultaneous users** submitting programs.
- Monitored **resource usage, execution time, and failures**.
- Fixed minor issues related to **memory leaks and timeout handling**.
- Documented backend reliability and performance metrics.

Technologies Used: Python, FastAPI, Docker, WSL2

Outcome:

Backend is now **highly reliable**, can handle multiple simultaneous executions without failure.